

Threshold Selection and Its Impact on F1-Score in Autoencoder-Based Anomaly Detection

1 Introduction

In this work, an autoencoder model is used to detect anomalies in blood cell images based on reconstruction error measured by Mean Squared Error (MSE). The anomaly detection decision is made by comparing reconstruction error against a predefined threshold.

2 Initial Threshold and Observed Issue

A fixed threshold of:

$$\text{Threshold} = 0.001656 \tag{1}$$

was used to classify samples as:

- Normal if $\text{MSE} \leq \text{threshold}$
- Anomalous if $\text{MSE} > \text{threshold}$

Observed Results

- ROC-AUC = 1.000
- F1-score = 0.0286

3 Analysis of Low F1-Score

Despite achieving a perfect ROC-AUC, the F1-score was extremely low. This issue arises due to improper threshold selection.

3.1 Reconstruction Error Distribution

We have:

- Normal images MSE range:

$$0.00012056809 \leq \text{MSE} \leq 0.0033889648 \tag{2}$$

- Anomalous images MSE:

$$\text{MSE} = 0.024621127 \quad (3)$$

3.2 Key Observation

There exists a clear separation between normal and anomalous samples:

$$\max(\text{Normal}) = 0.0033889648 \quad (4)$$

$$\min(\text{Anomaly}) = 0.024621127 \quad (5)$$

4 Cause of False Positives

The chosen threshold:

$$0.001656 \quad (6)$$

lies within the normal data distribution rather than between normal and anomaly distributions.

Effect

- Many normal samples exceed the threshold
- These are incorrectly classified as anomalies (false positives)

Impact on Metrics

- Precision decreases
- Recall remains high
- F1-score decreases significantly

5 Why ROC-AUC Remains Perfect

ROC-AUC measures the ranking ability of the model rather than classification using a fixed threshold.

Since:

$$\text{MSE}_{\text{anomaly}} > \text{MSE}_{\text{normal}} \quad \forall \text{samples} \quad (7)$$

the ranking is perfect, resulting in:

$$\text{ROC-AUC} = 1.0 \quad (8)$$

6 Optimal Threshold Selection

To correctly separate normal and anomalous samples, the threshold must lie between the two distributions.

6.1 Threshold Calculation

$$\text{Threshold} = \frac{\max(\text{Normal}) + \min(\text{Anomaly})}{2} \quad (9)$$

Substituting the values:

$$\text{Threshold} = \frac{0.0033889648 + 0.024621127}{2} \quad (10)$$

$$\text{Threshold} \approx 0.014005 \quad (11)$$

7 Improved Classification Performance

Using the corrected threshold:

- All normal samples are correctly classified
- All anomalous samples are correctly classified

Expected Results

- F1-score ≈ 1.0
- No false positives
- No false negatives

8 Key Insights

- Threshold selection is critical in anomaly detection
- Thresholds are not transferable across datasets
- ROC-AUC alone is insufficient for classification evaluation
- Proper threshold selection significantly improves F1-score

9 Conclusion

The low F1-score was not due to poor model performance but due to an incorrectly chosen threshold. By analyzing reconstruction error distributions and selecting a threshold between normal and anomalous values, classification performance improves significantly, achieving near-perfect results.